

Chemistry

Unit 2

Practice Exam 1 Answers

Section one: Multiple-choice

Question 1

- A** The stronger the intermolecular forces between molecules in a liquid, the lower the vapour pressure. A comparison of boiling points provides an indication of the relative strength of intermolecular forces in a liquid. Water (H_2O) has the highest boiling point, indicating that it has the strongest intermolecular forces and hence the lowest vapour pressure.

Question 2

- C** Diatomic molecules, e.g. HF , can only be linear. H_2S is V-shaped. The sulfur atom has two bonding pairs of electrons and two non-bonding pairs. The four pairs of electrons take on a distorted tetrahedral arrangement that results in a V-shaped H-S-H structure. CH_4 has four C-H bonds in a tetrahedral structure.

Question 3

- D** HF , OF_2 and NF_3 are all polar molecules due to the polar nature of their bonds with fluorine. In CF_4 the polarity of the individual C-F bonds are cancelled out due to the symmetrical tetrahedral structure of the molecule.

Question 4

- C** HBr is a polar molecule and dipole-dipole attraction and dispersion forces operate between HBr molecules. Dipole-dipole bonds formed between molecules where hydrogen is bonded to fluorine, oxygen or nitrogen are referred to as hydrogen bonds. Dispersion forces, resulting from the attraction between instantaneous dipoles caused by the random movement of electrons, are also present in HBr .

Question 5

- B** For a given mass of a gas at constant temperature the volume is inversely proportional to the pressure, as represented in graph II.

Question 6

- A** There is a direct relationship between the absolute temperature and volume in a given mass of gas at a constant pressure. Such a relationship is represented in graph I.

Question 7

- A** At constant temperature and pressure there is a direct relationship between the amount, in moles, of a gas and its volume. Such a relationship is represented in graph I.

Question 8

- D** For a given mass of a gas at constant temperature, the volume is inversely proportional to pressure. The product of pressure and volume, pV , will be constant as shown in graph IV.

Question 9

- A** The molecules shown in B, C and D all contain polar bonds. These compounds are able to form hydrogen bonds, or dipole-dipole interactions, with water molecules and are consequently soluble. Compound A contains no polar bonds. The energy released by the formation of any dispersion forces between molecules of compound A and water is not sufficient to break the hydrogen bonds between water molecules and the dispersion forces in compound A. Consequently, the compound is least soluble in water.

Question 10

- D** The reaction with methyl red indicates that the pH is greater than 6.0. The reaction with bromothymol blue indicates a pH greater than 7.6. The reaction with phenolphthalein indicates a pH less than 8.2. From this data it can be deduced that the pH is between 7.6 and 8.2. Note that within the stated pH ranges indicators have an intermediate colour.

Question 11

- B** HCl is a strong acid and is completely dissociated. The HCN solution has the same concentration as the HCl solution but its pH is greater indicating a lower concentration of H^+ ions and that it is a weak acid. As a rule of thumb, concentrations less than 2 M are considered dilute and concentrations greater than 2 M are considered concentrated. 5 M HCN is a concentrated solution of a weak acid.

Question 12

- C** At higher temperatures, particles move faster than at lower temperatures and consequently their average kinetic energy is greater. More particles have energy greater than the activation energy and more collisions result in a successful reaction.

Question 13

- A** The enthalpy of reaction, ΔH , is the difference between the energy of the products and the energy of the reactants. The activation is the energy required to break bonds in the reactants and form the transition complex.

End of section one

Section two: Short answer

(Note: In multi-mark questions each allocated mark is indicated as * and total marks are indicated in parenthesis.)

Question 14

- a methane (1 mark)
- b chloromethane (1 mark)
- c fluorine, nitrogen, oxygen, carbon dioxide, ethene (1 mark)
- d fluorine, nitrogen, oxygen, carbon dioxide (1 mark)
- e ammonia, hydrogen fluoride (1 mark)
- f hydrogen sulfide (1 mark)

Question 15

- a mass alumina = 45% of 1.00 kg = 450.0 g
molar mass $\text{Al}_2\text{O}_3 = 2 \times 26.98 + 3 \times 16.00 = 101.96 \text{ g mol}^{-1}$ *
 $n(\text{Al}_2\text{O}_3) = \frac{450.0}{101.96} = 4.413 \text{ mol}^*$ (2 marks)
- b From the equation
 $n(\text{Al})/n(\text{Al}_2\text{O}_3) = 4/2 = 2$
 $n(\text{Al}) = 2 n(\text{Al}_2\text{O}_3)$
 $= 2 \times 4.413 = 8.826 \text{ mol}^*$
 $m(\text{Al}) = 8.826 \times 26.98 = 238.12 \text{ g}$
 $= 238 \text{ g (3 sig figs)}^*$ (2 marks)
- c From the equation
 $n(\text{CO}_2)/n(\text{Al}_2\text{O}_3) = 3/2 = 1.5$
 $n(\text{CO}_2) = 1.5 n(\text{Al}_2\text{O}_3)$
 $= 1.5 \times 4.413 = 6.62 \text{ mol}^*$
 $V(\text{CO}_2) = n(\text{CO}_2) \times 22.71$
 $= 6.62 \times 22.71 = 150.34$
 $= 150 \text{ L (3 sig figs)}^*$ (2 marks)

Question 16

- a silver nitrate (1 mark)
- b $\text{Ag}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{AgCl}(\text{s})$ (1 mark)
- c molar mass $\text{AgCl} = 143.35 \text{ g mol}^{-1}$
 $n(\text{AgCl}) = 2.65/143.35 = 0.01849 \text{ mol}^* = n(\text{AgNO}_3)$
 $c(\text{AgNO}_3) = n/V = 0.01849/0.010 = 1.849 \text{ mol L}^{-1}^*$
 $= 1.85 \text{ mol L}^{-1} \text{ (3 sig figs)}$ (2 marks)

Question 17

- a** Arrhenius defined an acid as a substance that contains hydrogen and ionises in water to form a hydrogen ion, H^+ . (1 mark)
- b i** Strong Arrhenius acids are completely ionised in aqueous solution, e.g. nitric acid (HNO_3), hydrochloric acid (HCl) and sulfuric acid (H_2SO_4). (1 mark)
- ii** Weak Arrhenius acids are partly ionised in aqueous solution, e.g. ethanoic acid (acetic acid, CH_3COOH). (1 mark)
- iii** $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$ (1 mark)
- c** Strong acids are strong electrolytes, that is they readily conduct an electric current. Strong acids are completely ionised in solution forming hydrogen ions and an anion*. Weak acids are weak electrolytes. They are poor conductors of electricity because they are only partly ionised in aqueous solution*. (2 marks)

Question 18 (One mark for a balanced equation with respect to the number of atoms, one mark for balanced charges.)

- a** $2\text{H}^+(\text{aq}) + \text{Mg}(\text{s}) \rightarrow \text{Mg}^{2+}(\text{aq}) + \text{H}_2(\text{g})$ (2 marks)
- b** $\text{H}^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{H}_2\text{O}(\text{l})$ (2 marks)
- c** $2\text{H}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$ (2 marks)

Question 19

- a** As the reaction between hydrochloric acid and calcium carbonate proceeds, the reactants are consumed and their concentration decreases. As the concentration decreases the number of collisions between reactant particles also decreases*. There are fewer collisions that have a favourable orientation and energy greater than the activation energy. Hence the number of collisions that result in a successful reaction decreases*. (2 marks)
- b** As the temperature increases the particles in a reaction mixture move faster and their kinetic energy increases*. More particles have energy greater than the activation energy resulting in an increased rate of reaction*. For every 10°C rise in temperature, the rate of reaction is approximately doubled. (2 marks)

Question 20

- a** $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{g})$ (1 mark)
- b** -890 kJ mol^{-1}
(one mark for correct value, one mark for correct sign and units) (2 marks)
- c** $+1970 \text{ kJ mol}^{-1}$
(one mark for correct value, one mark for correct sign and units) (2 marks)

End of section two

Section three: Extended answer

Question 21

- a** All types of chromatography involve a stationary phase* and a mobile phase*. (2 marks)
- b i** As the solvent moves up the TLC, the food dyes are continuously adsorbed onto the stationary phase and desorbed or dissolved back into the liquid phase*. Each of the dyes in the mixture differs in the extent to which they adsorb onto the stationary phase and dissolve back into the liquid phase. Since each of the dyes undergoes this process to a different extent they separate as the chromatogram develops. The dye that dissolves most readily in the liquid phase moved highest up the TLC sheet. The dye that bonds most strongly onto the stationary phase moves the least distance*. (2 marks)
- ii** The R_f value is the same for both yellow dyes (0.75). Given that the solvent and the stationary phase are the same it is likely that the yellow dyes are the same. (1 mark)
- iii** The R_f value is the same for both dyes but the solvent is different. Therefore it is not conclusive that the dyes are the same. (1 mark)
- iv** Under identical conditions, prepare chromatograms of the suspect dye and a sample of dye that is known to be uncontaminated. Calculate the R_f values of the components of each dye*. The sample is likely not contaminated if the same set of R_f values is obtained. However, if the two sets of R_f values are not identical it can be concluded that the suspect sample is contaminated*. (2 marks)
- c i** The attraction of pentane to the stationary phase is less than that of octane and decane. (1 mark)
- ii** Pentane. The component that was eluted first has the shortest retention time in the gas chromatogram. (1 mark)

Question 22

- a** 16 g of sodium chloride dissolving in 50 mL water is the same as 320 g of sodium chloride dissolving in 1 L of water. This is less than the solubility of sodium chloride therefore all 16 g will dissolve. (1 mark)
- b i** The solubility of potassium sulfate is 120 g L^{-1} . This equates to 6 g per 50 mL*. If 9 g of potassium sulfate is mixed with 50 mL of water, 6 g will dissolve and 3 g will remain undissolved*. (2 marks)
- ii** At 80°C the solubility of potassium sulfate is 10.5 g per 50 mL water. All the 9 g will dissolve at 80°C . (1 mark)
- c** Since graphite is insoluble none will dissolve. (1 mark)
- d** Molar mass of $\text{K}_2\text{SO}_4 = 174.3 \text{ g mol}^{-1}$ *
 $n(\text{K}_2\text{SO}_4) = 120/174.3 = 0.689 \text{ mol L}^{-1}$ (3 sig fig)* (2 marks)
- e i** Add the mixture to 50 mL water and heat to 80°C *. Both salts are soluble at this temperature but graphite is insoluble. The graphite can be separated by filtration of the hot solution. It should be dried in an oven*. Allow the filtrate to cool to 20°C *. Some of the potassium sulfate will precipitate as the solution is saturated with respect to this salt at 20°C . The precipitated potassium sulfate can be recovered by filtration. The filtrate should be dried in an oven to obtain a pure sample of potassium sulfate*. (4 marks)
- ii** Sodium chloride is highly soluble at both 20°C and 80°C and the solutions are unsaturated at both temperature so NaCl will not precipitate out of the solution*. The solution contains both sodium chloride and potassium nitrate, so a pure sample could not be obtained if the mixture was evaporated to dryness*. (2 marks)

- f i** $\text{K}_2\text{SO}_4(\text{s}) \rightarrow 2\text{K}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$ (1 mark)
- ii** When potassium sulfate is added to water the hydrogen bonds between some water molecules are broken* as the positive end of the polar water molecules are attracted to the negatively charged sulfate ions. The negative end of the polar water molecules are attracted to the positive potassium ions*. Ion-dipole forces are formed between the water molecules and the positive and negative ions*. The ionic bonds between potassium and sulfate ions at the surface of the solid are broken as these ions pull away from the surface of the ionic solid by movement of water molecules*. The ion-dipole attractions are strong enough to overcome the ionic attractions in potassium sulfate and the hydrogen bonds in water*. (5 marks)
- g** In pure ethanol the molecules between the ethanol atoms are held together by hydrogen bonds. In pure water the molecules are also held together by hydrogen bonds. When ethanol is mixed with water the hydrogen bonds* between ethanol molecules and the hydrogen bonds between water molecules are broken*. New hydrogen bonds are formed between ethanol and water molecules*. (3 marks)

Question 23

- a** Bottle A barium chloride*
Bottle B potassium carbonate*
Bottle C ammonium chloride* (3 marks)
- b** An acid will react with a carbonate salt to form carbon dioxide gas, a salt and water. Bubbles of gas will be observed in the reaction*. The production of carbon dioxide can be confirmed by bubbling the gas through limewater, which turns cloudy or milky when carbon dioxide is present.
 $\text{CO}_3^{2-}(\text{aq}) + 2\text{H}^+(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})^*$ (2 marks)
- c** The presence of ammonium chloride can be confirmed by adding some sodium hydroxide solution and heating the mixture*. Ammonia gas, which has a characteristic pungent odour, is produced. Ammonia gas will also turn a piece of red moist litmus paper blue.
 $\text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq}) \rightarrow \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l})^*$ (2 marks)
- d** Flame tests* could also be used to identify the contents of the bottles.
Barium chloride shows a yellow green flame colour*.
Potassium carbonate shows a lilac flame colour*.
Ammonium chloride has no characteristic flame colour but can be deduced by the process of elimination*. (4 marks)

End of answers